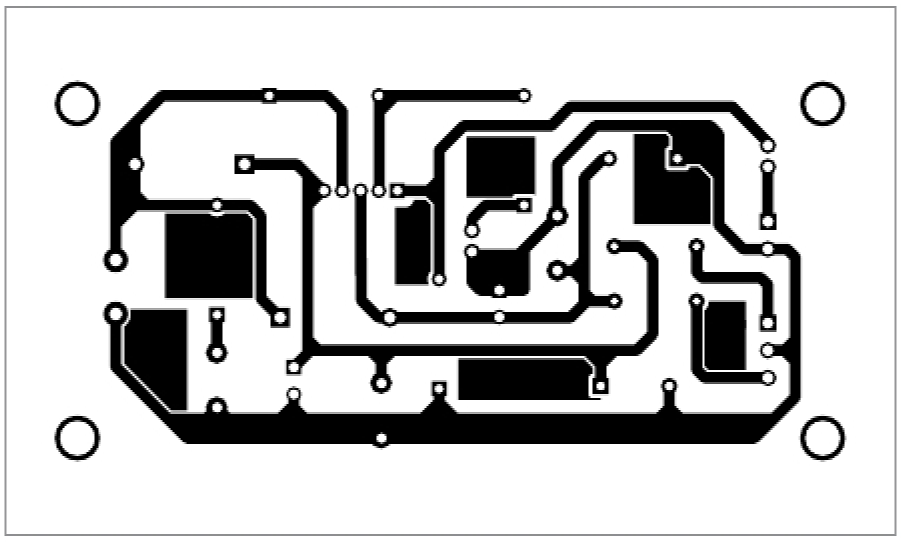

Fig. 3: Actual-size PCB layout of the amplifier

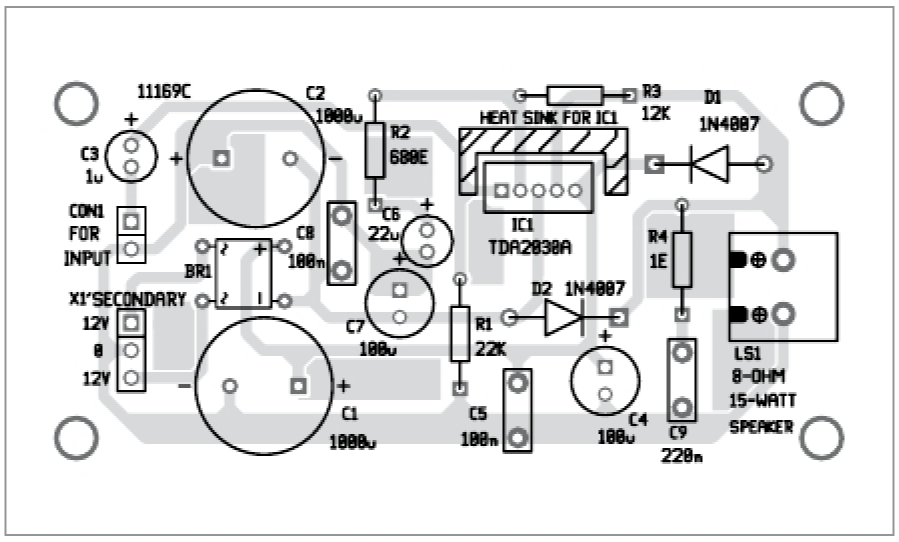

Fig. 4: Components layout of the PCB

laptop/mobile phone across CON1 and the 8-ohm, 15-watt speaker across LS1. Connect the transformer's secondary output (12V-0-12V) to the PCB and its primary to 230V AC mains.

You can also construct your amplifier on veroboard (5cm × 7cm). Mount IC1 in the middle of veroboard. Use a 2-pin connector on the left side and a 3-pin connector on the right side of the veroboard. The 2-pin connector is used for audio input, and the 3-pin connector is for dual power supply. Solder all the remaining components like resistors, capacitors, diodes, etc, using a low wattage soldering iron. You will need a suitable heat sink for IC1.

After connecting the dual power supply (12V-0-12V) to the circuit, you can test the circuit. For that, take a metal screwdriver and gently touch on the input terminal. You will hear a humming sound from the speaker, which means your amplifier is working. Note that the output power of amplifier will depend on the power supply.

Connect audio input at CON1 from any audio source such as MP3 player, laptop, or mobile phone. EFY

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